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1) If the entire quantity of all the reactants is not consumed in the reaction

which is the limiting reagent ?

2) Calculate the quantity of urea formed and unreacted quantity of the excess reagent.

The balanced equation is

$$2\ NH_3 + CO_2 \downarrow H_2NCONH_2 + H_2O$$

Answer :

1. The entire quantity of ammonia is consumed in the reaction. So ammonia is the limiting reagent. Some quantity of CO_2 remains unreacted, so CO_2 is the excess reagent.

2) Quantity of urea formed

= number of moles of urea formed × molar mass of urea

= 19 moles × 60 g mol^{-1}

= 1140 g = 1.14 kg

Excess reagent leftover at the end of the reaction is carbon dioxide.

Amount of carbon dioxide leftover

= number of moles of CO_2 left over × molar mass of CO_2

= 7 moles × 44 g mol^{-1}

= 308 g.

	Reactants		Products	
	NH_3	CO_2	Urea	H_2O
Stoichiometric coefficients	2	1	1	1
Number of moles of reactants allowed to react $N = \frac{\text{Mass}}{\text{Molar mass}}$	$\frac{646}{17}$ = 38 moles	$\frac{1144}{44}$ = 26 moles	–	–
Actual number of moles consumed during reaction Ratio (2 : 1)	38 moles	19 moles	–	–
No. of moles of product thus formed	–	–	19 moles	19 moles
No. of moles of reactant left at the end of the reaction	–	7 moles	–	–